Lab 3: Architectural Decision & Justification

Warehouse Inventory & Pallet Location Tracker System

Architecture Selection

We chose Microservices Architecture for the Warehouse Inventory & Pallet Location Tracker System.

Reason 1 — Independent scaling for a strict latency target

NFR-001 requires the system to resolve any SKU's bin coordinate across 100,000 bins in under 100 ms. As a dedicated microservice, the Inventory Location Service can be given its own in-memory index/cache and scaled horizontally during peak shift-change scanning, without over-provisioning slower, less latency-sensitive services such as movement logging.

Reason 2 — Fault isolation for a safety-critical check

FR-001 requires that a pallet placement is always validated against rack load capacity before storage is confirmed. Running the Pallet Placement Validator as its own service means a fault or slowdown in a non-critical component (e.g. the audit logger) cannot block or corrupt placement validation, so core warehouse operations keep running even if a peripheral service degrades.

Security Advantage

Each microservice exposes a minimal API surface and can enforce its own authentication/authorization boundary — for example, only the Pallet Placement Validator is granted write access to rack-capacity records. This least-privilege separation shrinks the blast radius of a compromised component compared with a single monolithic process holding full data access, directly supporting NFR-001's security requirement.

Performance Benefit

Because the Inventory Location Service is deployed and scaled independently, it can be backed by a dedicated caching layer and scaled out during peak load, letting it consistently meet the sub-100 ms lookup target without being throttled by slower, batch-oriented components like stock movement logging.

UML DIAGRAM

